

The Andrews–Curtis Conjecture



heuristic search for trivializations

Avigya Paudel · SURF 2026

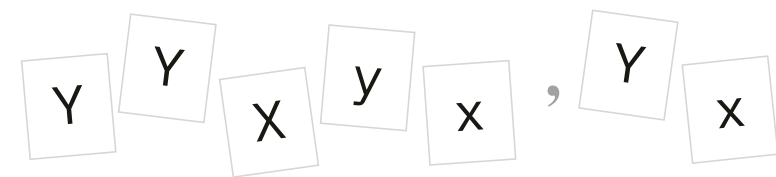
Prof. Sergei Gukov · Math AI Lab, Caltech

Lucas Fagan · Giorgi Butbaia · Michele Tarquini

Both are the trivial group

$$\langle x, y \mid x, y \rangle$$

the trivial group · obviously



the trivial group · provably

same group, different description

Three legal moves

AC moves · 1965

1

invert

$$r \rightarrow r^{-1}$$

x y → Y X

2

conjugate

$$r \rightarrow w r w^{-1}$$

x y → (x) x y (X)

3

multiply

$$r_1 \rightarrow r_1 r_2$$

x y · r₂

the group never changes

Two substitutions

$\langle YYXyx, Yx \rangle$

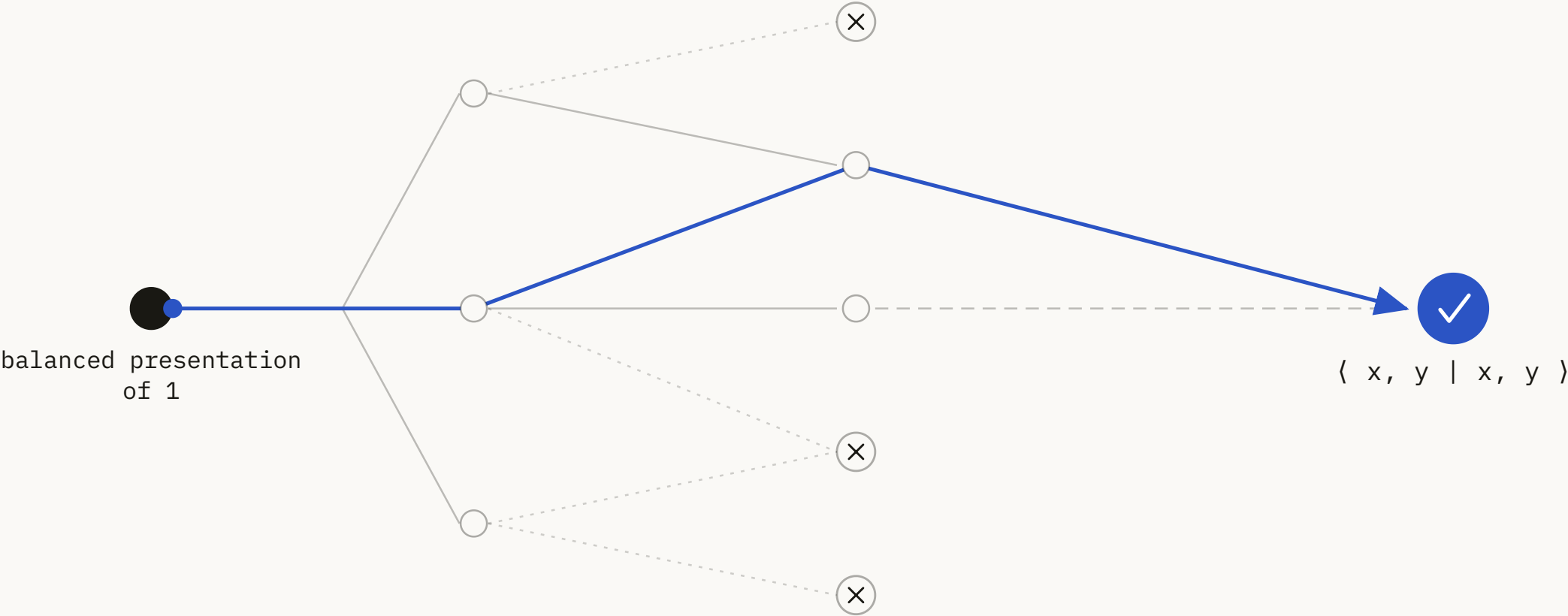


trivial



Does a path always exist?

Andrews-Curtis · open since 1965

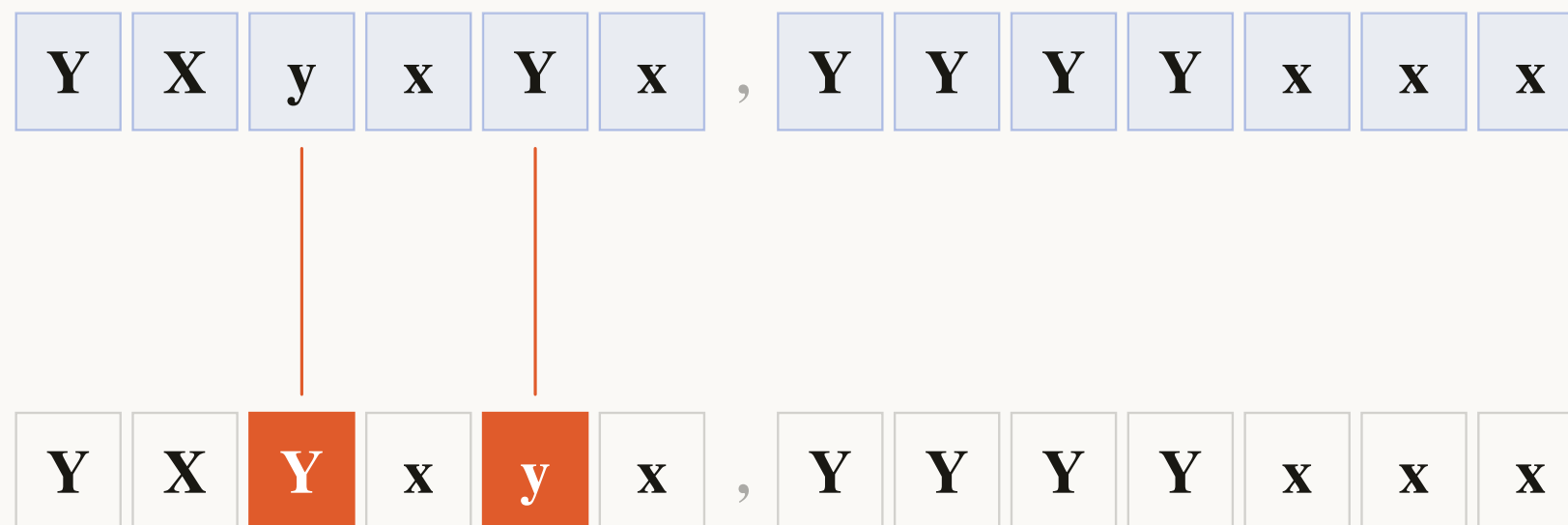


● a trivialization path exists?

⊗ dead ends everywhere

Two letters away from solved

AK(3) · Akbulut-Kirby 1985



solved · 12 moves

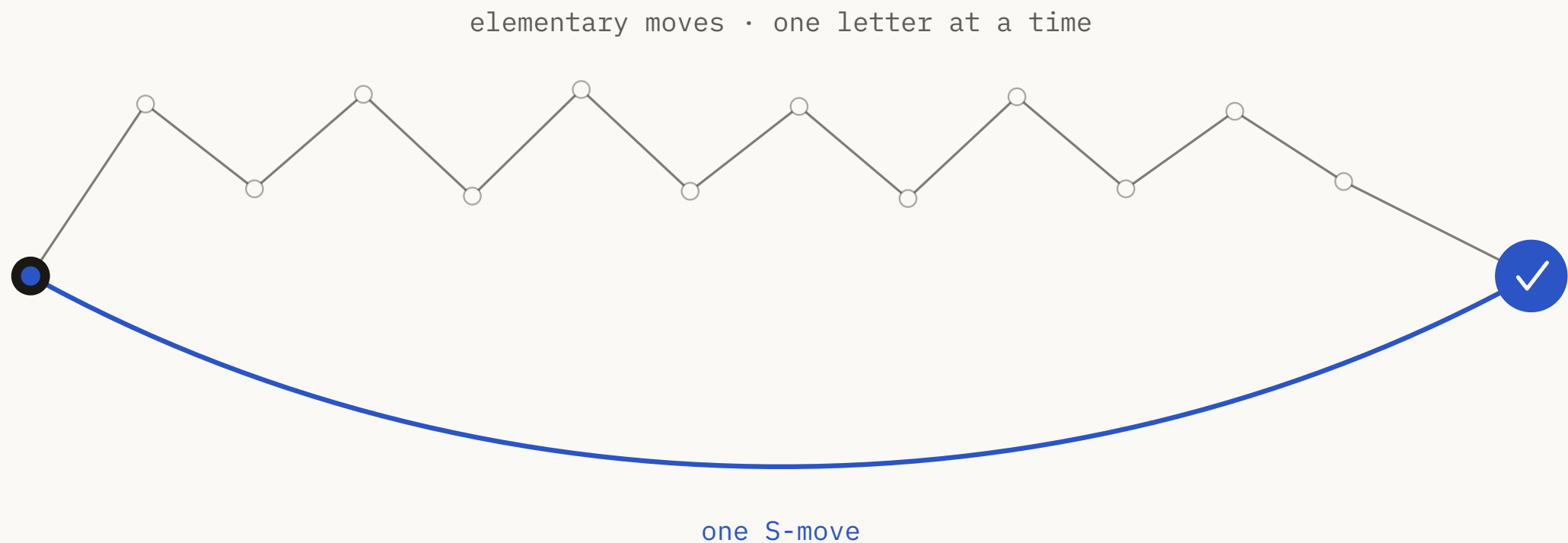


never solved

every known method fails on the bottom row

Many moves, one substitution

the S-move

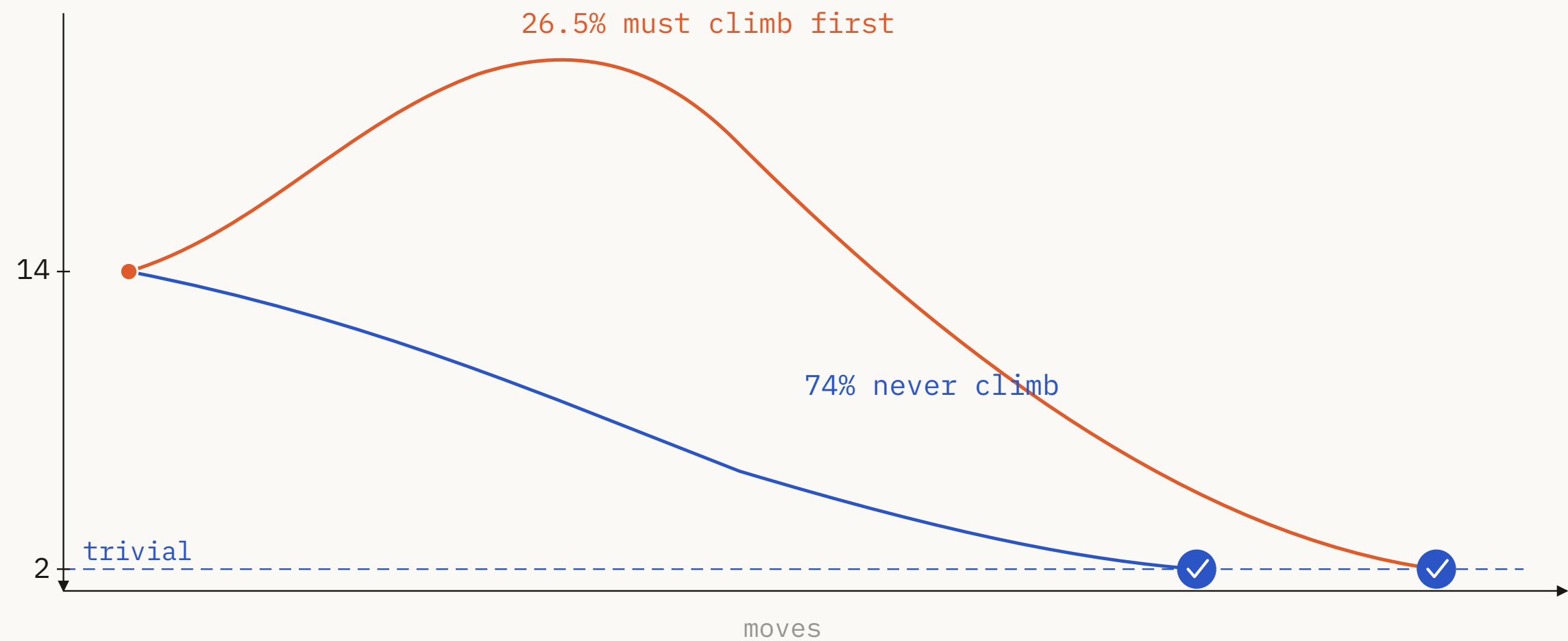


rotate r_1 · rotate & invert r_2 · concatenate · substitute

same reachable presentations · bigger strides per search node

Greedy search isn't enough

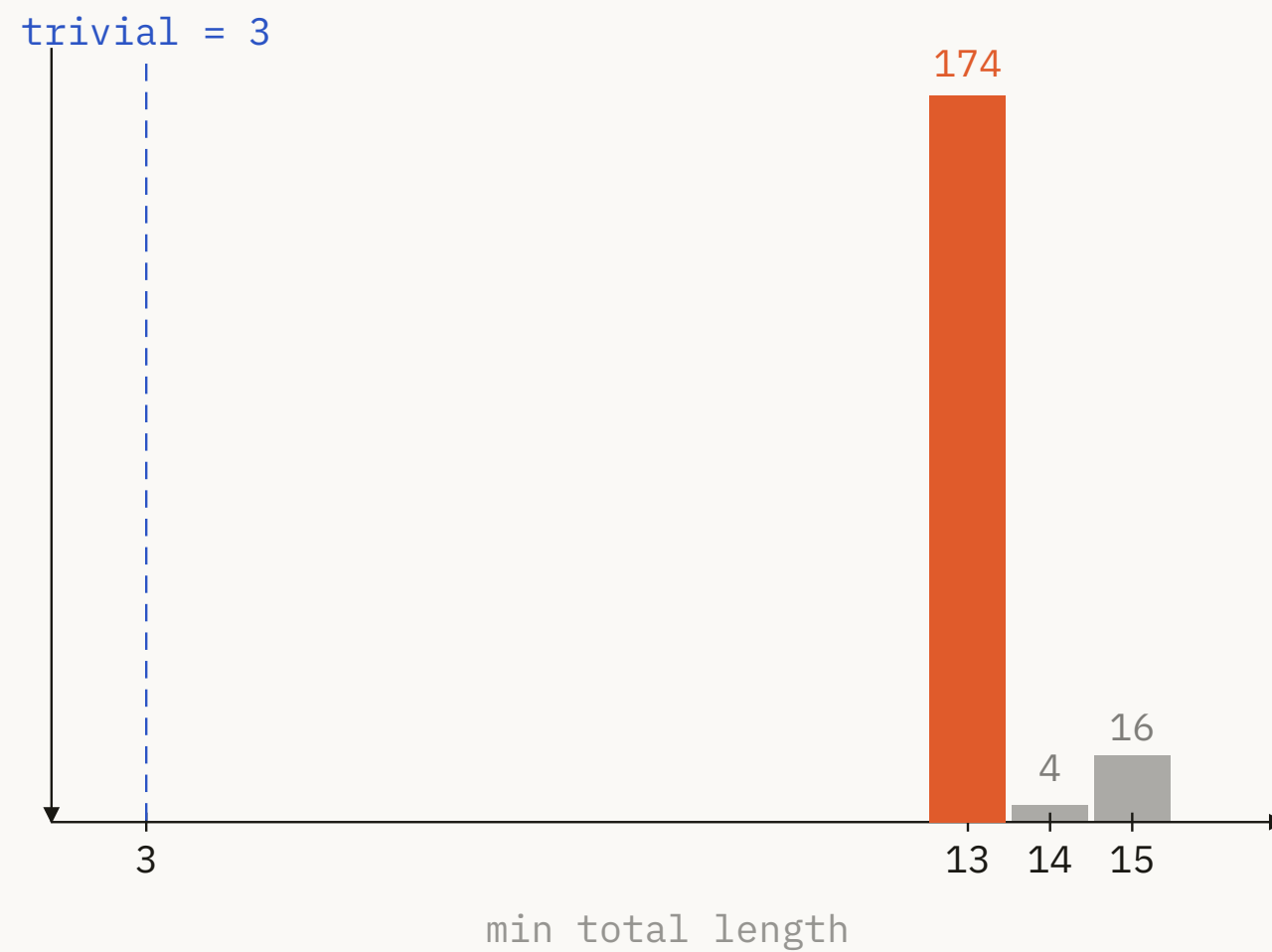
greedy on total length L



greedy always shortens – some solutions must get longer first

AK(3) floors at 13

all stabilization attempts



0 / 194

stabilizations solve

180,645

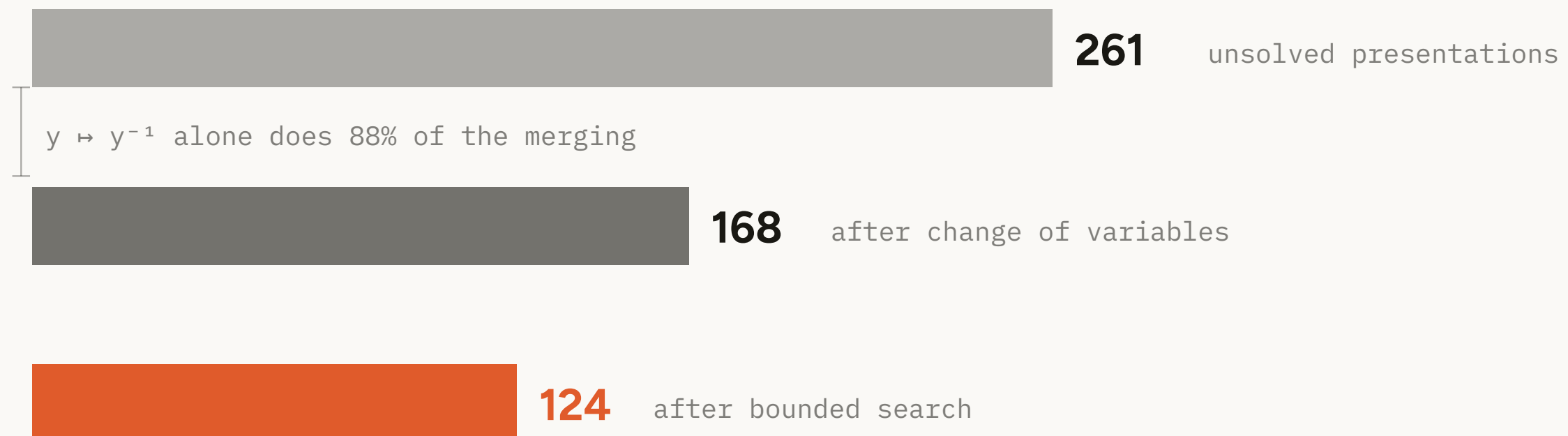
harvested quotients

16,870 → 0

re-solved · solved

261 problems are really 124

the unsolved set, deduplicated



Unsolved words have long blocks

S = mean run of the scarcer letter

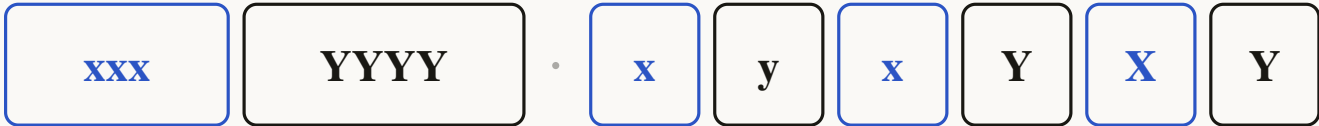
ms0



S = 1.00



AK(3)



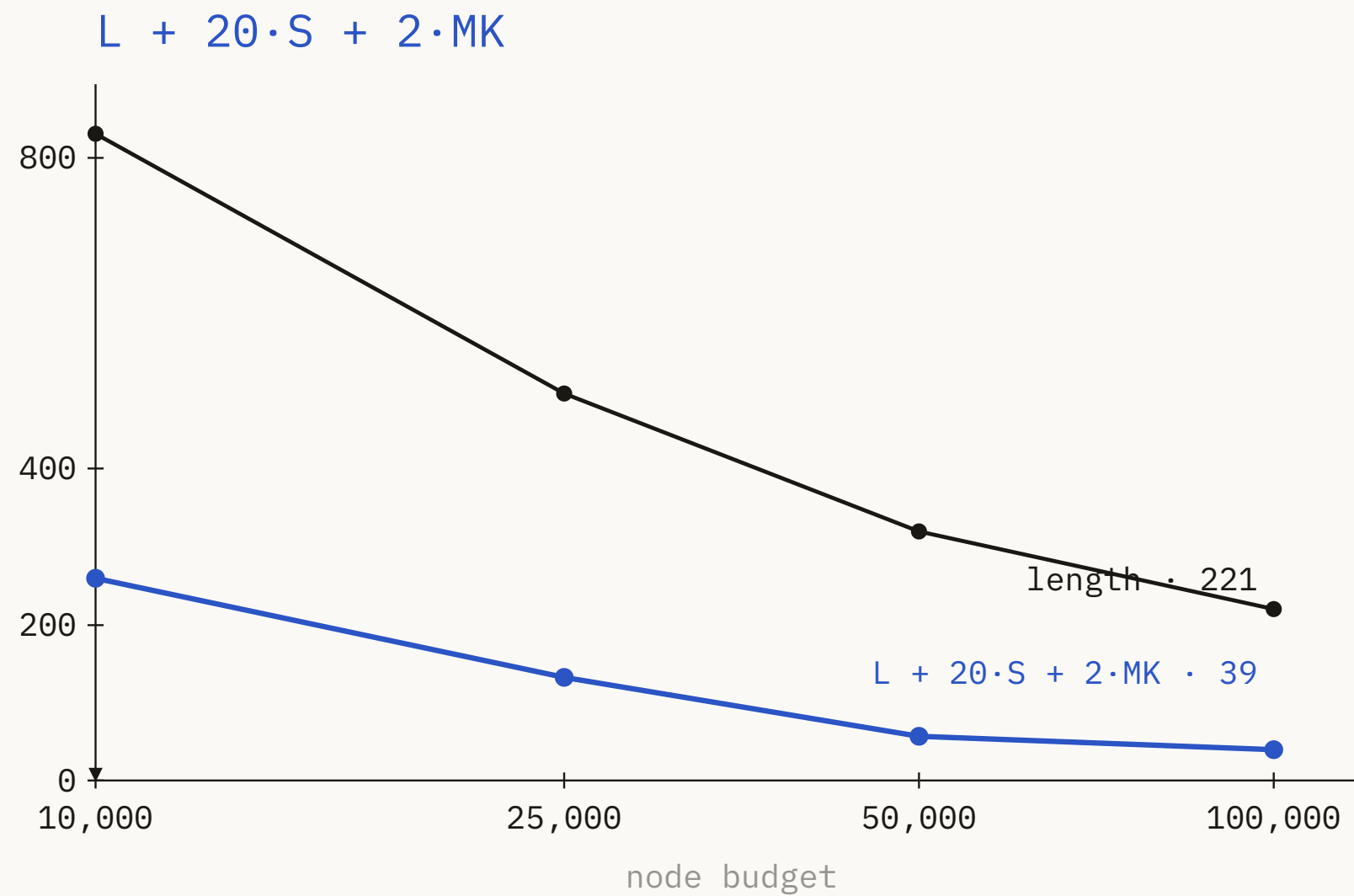
S = 1.50



S alone separates solved from unsolved · AUC 0.912

A better heuristic: 221 → 39

70,723 Aut-minimal orbits ·
budget 100,000



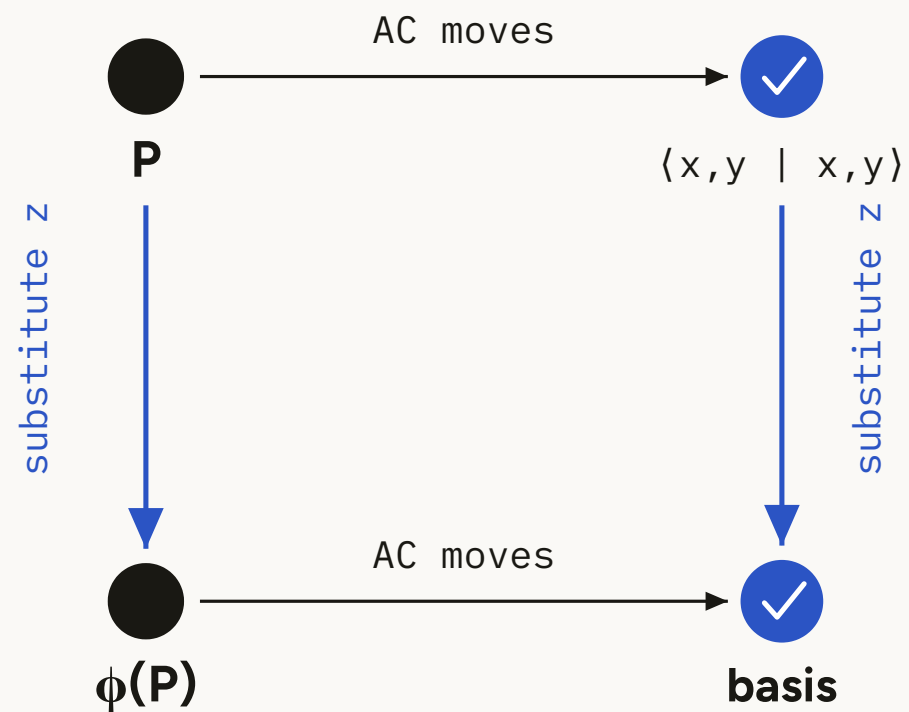
+182 -0
flips at 100k

95 → 277 / 316
hardest bin solved

563.3 → 263.2
mean nodes

Change the start, then search

change of variables



rewriting the problem is legal

52.2%

random CoV beats greedy – a coin flip

~109

candidate CoVs per presentation

45 / 60

oracle best-CoV (not a policy)

Mass picks the start

all 640 Miller-Schupp · budget
100,000

$$\text{mass} = |\sum x \ r_1| + |\sum y \ r_1| + |\sum x \ r_2| + |\sum y \ r_2|$$
 one pass · no search

plain greedy · 3,176,297 nodes · 634/640 at 100k



(640 needs 1M)

mass-ranked CoV, rank 1 · 420,426 nodes · 640/640

 **7.6×** fewer nodes

the six greedy never solved

☐ 634 · 73×

☐ 635 · 73×

☐ 636

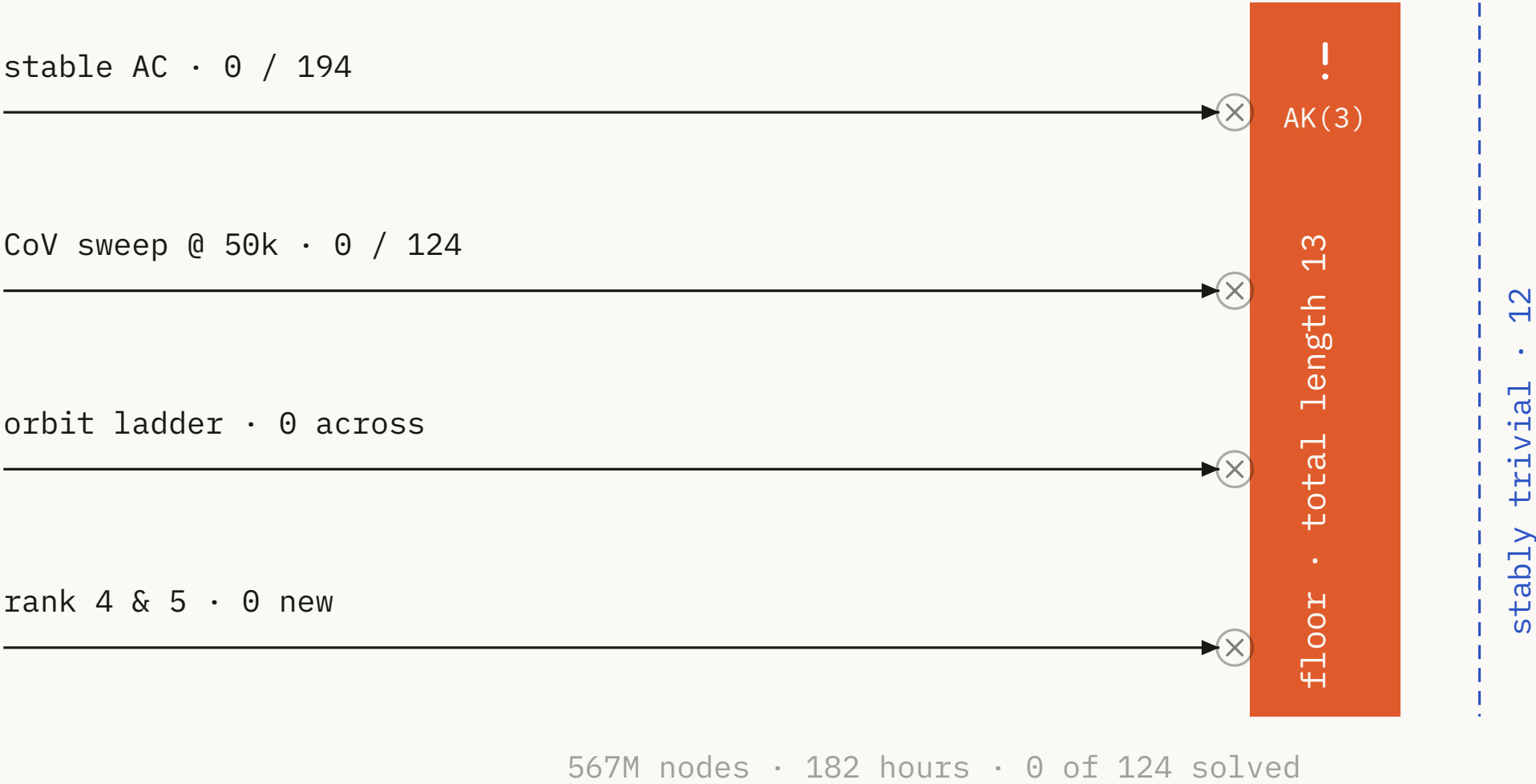
☐ 637

☐ 638

☐ 639

Still unsolved

124 classes · AK(3)



Two ideas solved all 640



what moves the floor?

Thank you

Prof. Sergei Gukov

Math AI Lab · Caltech

Lucas Fagan

Giorgi Butbaia

Michele Tarquini

mentors · Math AI Lab · Caltech

funded by Caltech Student-Faculty Programs · SURF 2026



Feedback

Summer Seminar Day 2026



Avigya Paudel

scan to send feedback

Appendix · the 7.6× needs early stopping

deployed (stop at first solve) · 420,426



as-run (all 3 ranks) · 8,447,944

greedy · 3,176,297



0

8,447,944

run rank 1, stop on solve

Appendix · cheaper nodes, longer paths

length · 23.5

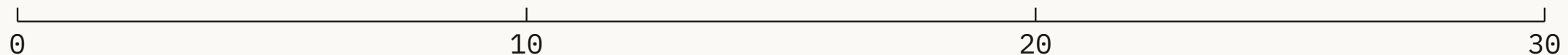


23.5

heuristic · 26.6



26.6



the heuristic spends moves to save nodes · 70,502 both-solved

Appendix · raising the cap doesn't help

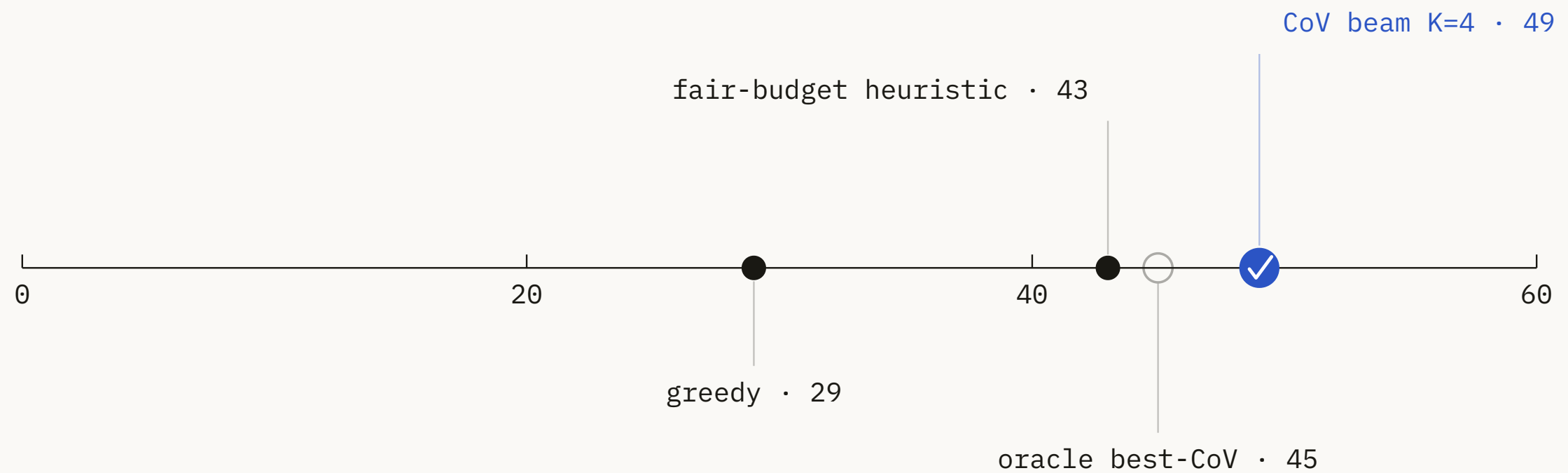
L = 24 ≡ L = 48

byte-identical visited sets · 16 / 16 runs

0 / 1,800

over-cap candidates solve

Appendix · a CoV beam beats the oracle

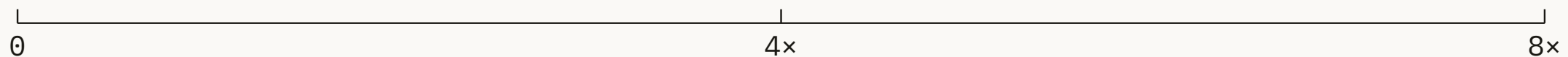


Appendix · relabelers inflated the CoV headline

all CoVs · median cut 7.58x



non-automorphic only · 3.62x



87% of CoV candidates are relabelings

Appendix · AK(3) in the literature



Appendix · how we know the group is 1

1

balanced

2 generators, 2 rules

2

abelianization

letter-counts · det ± 1

3

helper letter

add z → it trivializes

the group is 1

AC forbids the helper letter

AK(3) is stably trivial · still no AC path